

PhyST-Net: An Exogenous-Aware Spatio-Temporal Graph Transformer for Domain-Constrained Network Forecasting

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Abstract—Spatio-temporal forecasting is a core task for graph-structured networks such as power grids, traffic systems, and industrial sensor clusters, where accurate node-level prediction requires joint modeling of historical trends, external drivers, and operational constraints. Existing forecasting models often treat historical covariates, future exogenous variables, and domain priors as isolated inputs, limiting their ability to capture the interplay between temporal continuity, dynamic spatial topology, and physical feasibility. This paper proposes PhyST-Net, an exogenous-aware spatio-temporal graph Transformer designed as a general framework for domain-constrained network forecasting. PhyST-Net formulates prediction as a multi-modal graph learning task with historical target states, historical exogenous variables (ExG), future ExG, and graph metadata as distinct interfaces. The framework employs: (1) Adaptive Gaussian-soft-windowed Patching (AGSP) for continuous temporal tokenization; (2) a Relational Spatio-Temporal Manifold Fusion (R-STMF) graph adapter for multi-scale spatial dependency modeling; (3) KAN-enhanced Adaptive Conditioning (K-AC) for non-linear exogenous modulation; and (4) a Domain-Informed Dynamic Capping Head (DI-DCH) for constraint-aware decoding. While the framework is general, we validate its effectiveness on the challenging task of wind power forecasting using the SD-WPF, HoT, and Kelmarsh datasets. Results show that PhyST-Net consistently improves forecasting accuracy and feasibility across multiple horizons, demonstrating the value of explicit exogenous awareness and domain-informed architectural inductive biases.

Index Terms—Spatio-temporal forecasting, graph Transformers, exogenous variables, domain-constrained machine learning, network forecasting.

I. INTRODUCTION

SPATIO-TEMPORAL forecasting aims to predict future node states in graph-structured systems where temporal dynamics and spatial interactions evolve jointly, such as transportation networks, power grids, and industrial sensor clusters. In these applications, node-level trajectories are rarely self-driven; rather, they are shaped by a combination of historical operating conditions, future external drivers, and domain-specific feasibility constraints. As illustrated in Fig. 1, complex network signals typically comprise shared macro-trends driven by exogenous forcing and high-frequency variations unique to individual nodes. Consequently, a practical forecasting model must learn continuous temporal representations, infer dynamic graph relations, and integrate future exogenous variables (ExG) without inducing data leakage, all while ensuring that its predictions remain compatible with operational regimes.

A. The Forecasting Trilemma in Spatio-Temporal Networks

Despite recent architectural advancements, modern spatio-temporal forecasting frameworks remain hampered by a “Fore-

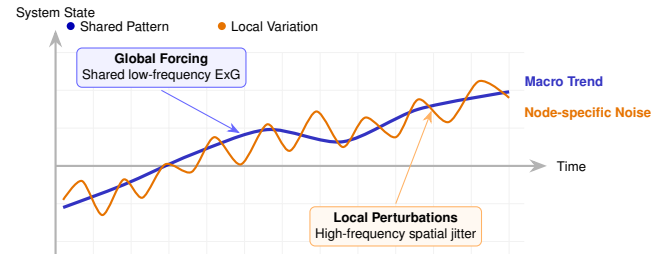


Fig. 1. Multi-scale spatio-temporal processes. General network signals are composed of shared macro-trends driven by exogenous forcing and high-frequency variations unique to individual nodes. PhyST-Net separates these components through dual-branch graph reasoning.

casting Trilemma”—an inherent trade-off between Model Capacity, Domain Consistency, and Spatio-Temporal Complexity.

1) *Model Capacity vs. High-Frequency Noise*: Capacity denotes the expressive power of the underlying mathematical architecture. While deep learning models, particularly Transformers, possess immense expressive capability for sequence modeling, they are notoriously sensitive to the high-frequency stochastic noise embedded in raw spatio-temporal data (e.g., localized traffic congestions or turbulent wind gusts). Furthermore, standard temporal patching exacerbates this vulnerability by splitting continuous physical regimes across arbitrary boundaries. As highlighted by recent advancements in complexity-balanced sequence modeling, deploying uniform representational capacity across an entire time horizon is computationally inefficient and structurally rigid. The central challenge is designing an architecture equipped with sufficient capacity to capture macro-trend signals, while concurrently employing adaptive temporal allocation to filter node-specific noise.

2) *Domain Consistency and Constraint-Aware Operations*: Consistency mandates that a model’s outputs adhere to the fundamental physical laws or operational limits inherent to the target domain. Purely data-driven architectures frequently generate unfeasible predictions—such as negative vehicular speeds or power generation exceeding a facility’s nameplate capacity. Such violations severely degrade operator trust in critical infrastructure. Inspired by emerging paradigms in safety-critical generation that utilize output-space guidance, a robust forecasting framework must enforce physical feasibility directly within its decoding pathway, moving beyond the limitations of reactive, post-hoc corrections.

3) *Spatio-Temporal Complexity and Multi-Scale Dynamics*: Complexity involves the management of dense spatial couplings across heterogeneous nodes. In physical networks, nodes interact via multiple simultaneous mechanisms: local

topological dependencies (e.g., aerodynamic wake effects, adjacent road segments) and global environmental fronts (e.g., synoptic weather patterns). Existing models frequently conflate these multi-scale interactions into a singular spatial operator, thereby failing to distinguish localized perturbations from system-wide drivers.

To systematically resolve this trilemma, we propose PhyST-Net, an ExG-aware Spatio-Temporal Graph Transformer designed as a versatile framework for domain-constrained network forecasting. To accommodate the multi-scale and multi-modal nature of dynamic network signals, we introduce the following key innovations:

- **Continuous and Differentiable Patching:** We present the **Adaptive Gaussian-soft-windowed Patching (AGSP)** mechanism. By transforming discrete observations into learnable spectral-temporal densities, AGSP recovers continuous event dynamics and functions as an adaptive low-pass filter, effectively rejecting high-frequency noise.
- **Decoupled Spatio-Temporal Topology:** We introduce the **Relational Spatio-Temporal Manifold Fusion (R-STMF)** framework. R-STMF leverages a dual-branch graph logic to decompose latent node dynamics into multi-scale spatial manifolds, mitigating the spatial conflation typical of standard distance-based graphs.
- **Piecewise Conditioning and Safe Capping:** We implement the **KAN-enhanced Adaptive Meteorological Conditioning (K-AMC)** module, utilizing Kolmogorov-Arnold spline edge functions for the non-linear modulation of graph-enhanced states via exogenous drivers. This is coupled with a **Physics-Informed Dynamic Capping Head (PI-DCH)** that embeds strict physical safety constraints and adherence to operational envelopes directly into the output space.

By explicitly partitioning the data flow into distinct computational pathways—historical context, relational reasoning, and exogenous conditioning—PhyST-Net establishes a rigorous forecasting contract. This interface-centric design prevents the entanglement of historical data and future drivers, ensuring the architecture can be seamlessly instantiated across diverse critical domains.

B. Contributions

The primary contributions of this work are summarized as follows:

- **A general ExG-aware graph-forecasting formulation.** We define network forecasting as a multi-modal task where historical states, exogenous variables, and domain constraints are processed through distinguishable pathways, thereby supporting broad architectural generalizability.
- **A continuous temporal tokenization module.** We introduce AGSP, which supersedes rigid patch boundaries with learnable Gaussian soft-windows, preserving event continuity and minimizing boundary artifacts in non-stationary sequences.

- **A modular graph adapter for mixed spatial mechanisms.** We develop R-STMF to disentangle multi-scale spatial dependencies prior to relational propagation, providing a modular contract that accommodates alternative graph predictors.
- **A future-ExG-conditioned decoding path.** We fuse K-AMC and PI-DCH to inject forecast-horizon drivers and structurally enforce feasibility constraints within the model, bridging latent representation learning with strict operational requirements.

The effectiveness and robustness of PhyST-Net are validated through comprehensive experiments on real-world datasets spanning diverse scales and operational conditions, confirming its viability as a state-of-the-art general-purpose network forecasting framework.

II. RELATED WORK

This section reviews representative research trajectories pertinent to PhyST-Net. Rather than chronicling models chronologically, we categorize prior literature based on the core modeling challenges inherent to ExG-aware spatio-temporal forecasting: temporal dependency modeling, spatial graph learning, exogenous-variable fusion, and constraint-aware output generation.

A. Temporal Forecasting and Transformer Architectures

Classical statistical forecasting techniques, including autoregressive and state-space models, retain utility when predicting low-dimensional, approximately stationary processes. However, these foundational assumptions become restrictively fragile in graph-structured systems where node dynamics are highly non-linear and persistently coupled with external forcing. While deep recurrent and convolutional architectures improved non-linear sequence modeling via latent state tracking and local temporal filtering, long-horizon forecasting remains challenging due to error accumulation and vanishing gradients over extended time steps.

Transformer-based forecasters resolve long-range dependency constraints via self-attention, establishing themselves as the dominant paradigm for deep time-series analysis. Recent variants, utilizing PatchTST-style patching and inverted-dimension attention, significantly reduce computational overhead while enhancing representation learning for multivariate sequences [1], [5], [6]. Nevertheless, temporal patching is predominantly implemented via rigid, fixed-length windows. In highly non-stationary sequences, physical regime transitions and rapid fluctuations rarely align with these artificial boundaries. PhyST-Net advances this trajectory by introducing AGSP, rendering temporal tokenization fully learnable and mathematically continuous.

B. Spatio-Temporal Graph Forecasting

Spatio-temporal graph neural networks (STGNNs) capture node interactions via graph convolutions, diffusion propagation, and spatial attention [9], [10], [14]. These methodologies are highly effective when cross-node dependencies dictate

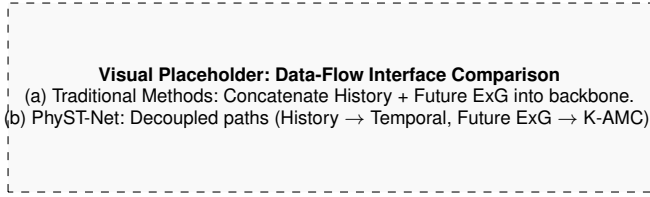


Fig. 2. Comparison of Exogenous Variable Fusion Interfaces. (a) Standard forecasting models frequently concatenate historical data and future drivers early in the network, increasing the risk of temporal leakage and confounding latent representation learning. (b) PhyST-Net explicitly decouples these data flows, reserving future drivers exclusively for horizon-level conditioning via the K-AMC module.

system behavior, such as in traffic corridors and power grids. Furthermore, adaptive graph learning relaxes the restrictive assumption of a static, fully observable topology, allowing models to infer latent relations directly from data and dynamically adjust propagation patterns.

A critical limitation, however, is the widespread reliance on homogeneous propagation protocols across all latent states. In physical systems, spatial dependence is inherently multi-modal; nodes interact through localized proximities, shared environmental influences, and overarching exogenous drivers. A singular adjacency matrix, even if learned dynamically, conflates these distinct mechanisms. R-STMF resolves this by decomposing temporal representations into high-frequency local variations and low-frequency macro-trends, applying targeted, sparse relational propagation to each manifold prior to fusion.

C. Exogenous-Variable-Aware Forecasting

Real-world forecasting tasks are intrinsically dependent on exogenous (ExG) variables, such as meteorological forecasts, control schedules, and market indices. Existing predictive models primarily integrate external data through direct feature concatenation, cross-attention mechanisms, or graph-enhanced fusion [26], [27]. While these mechanisms improve accuracy under ideal conditions, they frequently blur the temporal distinction between historical observations and future drivers.

Specifically, historical ExG elucidates recent state trajectories, whereas future ExG provides horizon-level modulation. Merging both into a singular feature tensor complicates data availability reasoning at inference time. Consequently, PhyST-Net structurally decouples historical ExG from future ExG (as illustrated in Fig. 2). K-AMC leverages the future-ExG pathway to generate latent modulation parameters, enabling forecast-horizon drivers to dynamically condition graph-enhanced states without being erroneously treated as historical priors.

D. Interpretable Conditional Modeling with KANs

Interpretability is paramount in operational forecasting, as dispatchers require insight into the primary drivers shaping a prediction. While attention weights and post-hoc attribution methods offer diagnostic utility, they often fail to reflect the model’s actual computational pathways. Kolmogorov-Arnold

Networks (KAN) [15] represent a paradigm shift from traditional perceptrons by modeling transformations via learnable, univariate spline-based edge functions [16]–[19].

By structuring a conditioning network as an additive sum of feature-wise edge functions, the influence of individual input variables can be analytically inspected through learned basis coefficients and derivatives. PhyST-Net capitalizes on this property within the K-AMC module: future ExG features are mapped to scale and shift parameters via spline functions. While not establishing strict causal identifiability, this architectural alignment provides a transparent, inspectable explanation mechanism far superior to generic dense concatenation.

E. Scalable Spatio-Temporal Forecasting

Computational scalability is a persistent bottleneck in graph forecasting. Dense spatio-temporal attention scales quadratically with both look-back length and graph cardinality. Conventional mitigation strategies—such as temporal down-sampling, rigid patching, and aggressive graph truncation—improve efficiency but risk discarding critical physical dynamics.

PhyST-Net approaches scalability dualistically. First, AGSP compresses the raw temporal dimension into a highly dense sequence of continuous tokens, preserving soft event boundaries while drastically lowering attention overhead. Second, R-STMF employs a sparse Top- K graph propagation scheme, ensuring that node interaction costs scale relative to a parameterized neighborhood size rather than the complete graph matrix. The theoretical complexity analysis detailed in Section IV-G connects these design choices directly to practical deployment constraints.

F. Physics-Informed and Constraint-Aware Forecasting

Physics-informed learning embeds essential domain knowledge into neural architectures via structural priors, penalty losses, or projection operators. In network forecasting, enforcing such constraints is vital, as system targets are strictly bounded by operational capacities. Unfeasible predictions, regardless of aggregate error metrics, can trigger catastrophic downstream decisions.

Contemporary constraint-aware frameworks predominantly rely on physics-guided loss regularization [21], [24] or reactive post-hoc clipping [20], [25]. However, soft penalties do not guarantee compliance, and post-hoc clipping creates a structural disconnect between the trained model and inference behavior. Aligning with the philosophy of structural physics induction [22], the PI-DCH layer internalizes feasibility operations directly within the forecasting head. By utilizing smooth barrier functions to shape predictions toward a dynamic capacity envelope prior to a final non-linear projection, the constraint pathway becomes an explicit, differentiable component of the computational graph.

G. Position of This Work

PhyST-Net synthesizes advancements in Transformer forecasting, adaptive graph learning, and interpretable exogenous

conditioning. Its distinguishing factor lies in the rigorous organization of the forecasting interface. Rather than presenting a bespoke, single-domain architecture, this paper formulates a generalized ExG-aware graph prediction framework, subsequently instantiating physically meaningful, domain-specific modules. This methodology ensures PhyST-Net serves both as a state-of-the-art general-purpose model and as a robust, reusable template for domain-constrained network forecasting.

III. PROBLEM DEFINITION

We formulate the task as ExG-aware spatio-temporal graph forecasting. Let $G = (V, E, A)$ denote a graph, where $V = \{v_1, \dots, v_N\}$ is the node set, E is the edge set, and $A \in \mathbb{R}^{N \times N}$ is an adjacency or relation matrix. A node can represent a network element, road sensor, grid bus, monitoring station, or industrial device depending on the application. The graph relation A may be physical, geographic, statistical, learned, or a combination of these sources.

At forecasting origin t , the historical target tensor is

$$\mathbf{X}_{t-T+1:t} \in \mathbb{R}^{T \times N \times D_x}, \quad (1)$$

where T is the look-back length, N is the number of nodes, and D_x is the target-state dimension. The future prediction target is

$$\mathbf{Y}_{t+1:t+H} \in \mathbb{R}^{H \times N \times D_y}, \quad (2)$$

where H is the forecast horizon and D_y is the output dimension.

We distinguish two types of exogenous variables. Historical ExG is available over the look-back window:

$$\mathbf{E}_{t-T+1:t}^h \in \mathbb{R}^{T \times D_h}, \quad (3)$$

and describes external or contextual conditions observed together with the target history. Future ExG is available over the forecast horizon:

$$\mathbf{E}_{t+1:t+H}^f \in \mathbb{R}^{H \times D_f}, \quad (4)$$

and represents known or forecastable future drivers. This separation is important because historical ExG estimates the current latent state, whereas future ExG conditions the unknown horizon. In spatio-temporal node prediction, historical ExG may include external variables and time features, while future ExG may include exogenous forecasts.

The formulation assumes an operational forecasting setting. At time t , the model has access to historical target states and historical ExG up to t , and it has access to future ExG for $t+1$ to $t+H$ only if those variables are known, scheduled, or forecast by an external source. The future target values are never used as inputs. This assumption is important for preventing leakage: variables that are measured only after the target horizon should not be included in $\mathbf{E}_{t+1:t+H}^f$. If an application lacks reliable future ExG, the same formulation can be used by setting $\mathbf{E}^f$ to calendar features, scenario forecasts, or a learned uncertainty representation, but the resulting claim should be stated as a different forecasting setting.

The graph relation is also allowed to be partially observed. In some systems, A is a physical adjacency matrix; in others,

it is a distance kernel, a correlation prior, or an empty prior that is refined by adaptive learning. We therefore separate A from M_G . The matrix A specifies direct relation priors used by graph propagation, while M_G collects auxiliary metadata such as node coordinates, node attributes, or environmental descriptors. This separation lets the same method support both fixed-topology and adaptive-topology forecasting.

The forecasting model is a parameterized function f_θ :

$$\hat{\mathbf{Y}}_{t+1:t+H} = f_\theta \left(\mathbf{X}_{t-T+1:t}, \mathbf{E}_{t-T+1:t}^h, \mathbf{E}_{t+1:t+H}^f, A, M_G \right), \quad (5)$$

where M_G denotes graph metadata such as coordinates, node attributes, physical distances, or topology priors. The model should satisfy three requirements. First, it should extract temporal representations from non-stationary node histories. Second, it should propagate information across graph nodes without assuming that all spatial dependencies share one mechanism. Third, it should use future ExG only through variables available at inference time.

When domain constraints are available, the model can be decomposed into an unconstrained spatio-temporal backbone $\mathcal{H}_\theta$ and a constraint operator $\mathcal{C}_\Omega$:

$$\hat{\mathbf{Y}}_{t+1:t+H} = \mathcal{C}_\Omega \left(\mathcal{H}_\theta \left(\mathbf{X}_{t-T+1:t}, \mathbf{E}_{t-T+1:t}^h, \mathbf{E}_{t+1:t+H}^f, A, M_G \right) \right), \quad (6)$$

where Ω denotes domain knowledge. In spatio-temporal node prediction, Ω includes non-negativity, capacity limits, and regime feasibility. In other domains, Ω may encode speed limits, conservation constraints, bounded sensor ranges, or operational safety limits.

Given a training set $\mathcal{D}$, the supervised objective is

$$\min_{\theta} \frac{1}{|\mathcal{D}|} \sum_{(\mathbf{X}, \mathbf{E}^h, \mathbf{E}^f, A, M_G, \mathbf{Y}) \in \mathcal{D}} \ell(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda_\Omega \mathcal{R}_\Omega(\hat{\mathbf{Y}}, \mathbf{Y}), \quad (7)$$

where $\ell(\cdot)$ is a forecasting loss, $\mathcal{R}_\Omega$ is an optional constraint-aware regularizer, and λ_Ω controls its strength. The formulation does not require a specific graph operator; any module that maps temporal node states to graph-enhanced node states can be used as the graph adapter.

IV. METHOD

PhyST-Net is an ExG-aware spatio-temporal graph Transformer designed for network-structured node forecasting with known external drivers and domain constraints. Given the inputs defined in Section III, the model follows the pipeline in Fig. 3. Historical target states and historical ExG are first embedded into node-level temporal representations. AGSP converts the temporal history into continuous tokens. A Transformer backbone learns temporal dependencies. R-STMF serves as a graph adapter that maps temporal node states to graph-enhanced node states. K-AMC injects future ExG into the latent representation. PI-DCH finally decodes the conditioned states into feasible forecasts respecting domain-specific bounds.

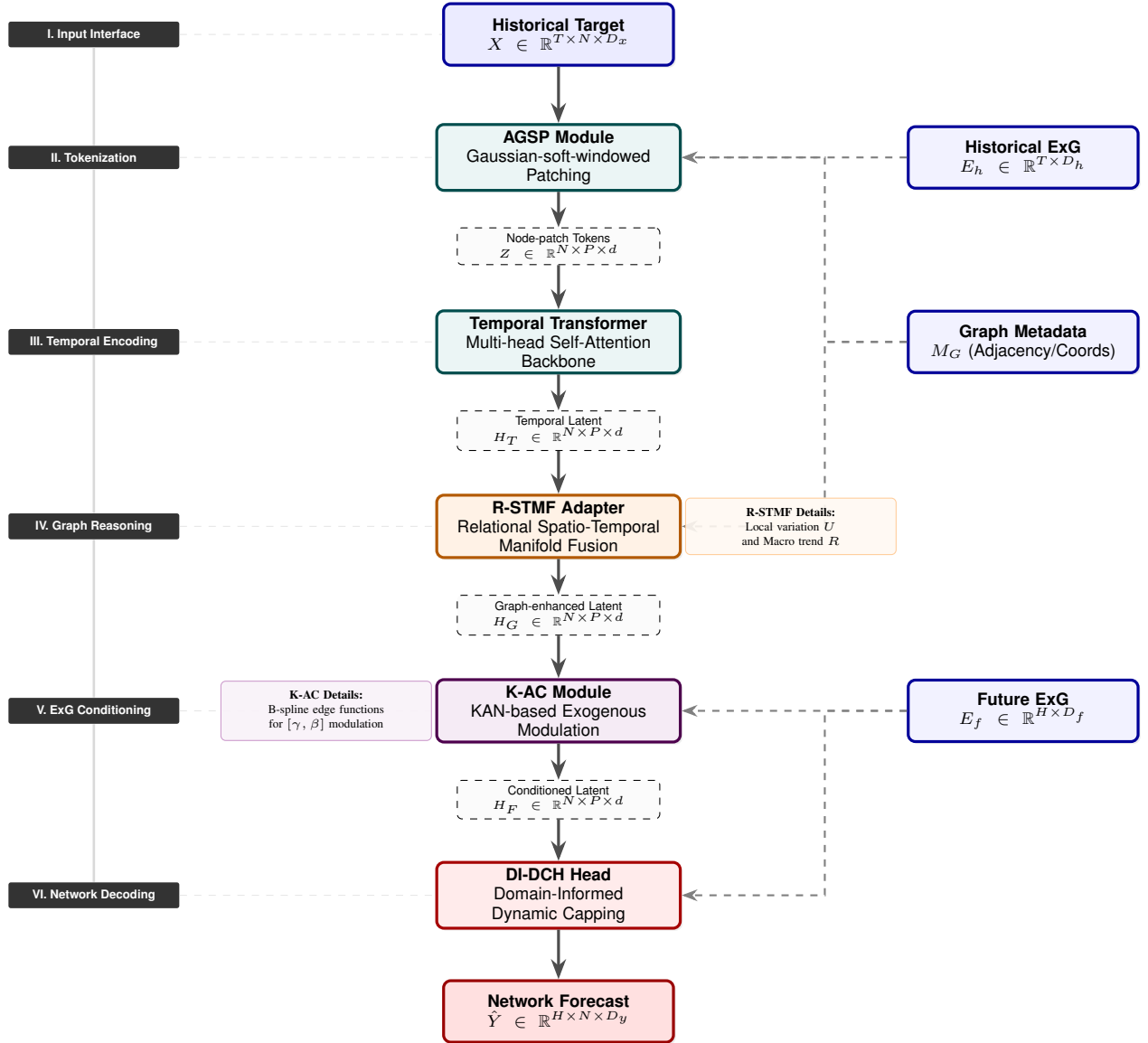


Fig. 3. Overall architecture of PhyST-Net. The framework operates as an exogenous-aware vertical pipeline: (I) Multi-modal input interface; (II) AGSP continuous tokenization; (III) Transformer sequence encoding; (IV) R-STMF relational manifold fusion; (V) K-AC spline-based conditioning; and (VI) DI-DCH domain-constrained forecasting. All external data (ExG and Metadata) and technical details are organized in peripheral tracks to ensure a clear and interpretable data flow.

A. Overall Framework

For a mini-batch, let $\mathbf{X} \in \mathbb{R}^{B \times T \times N \times D_x}$ denote historical target states, $\mathbf{E}^h \in \mathbb{R}^{B \times T \times D_h}$ denote historical ExG, and $\mathbf{E}^f \in \mathbb{R}^{B \times H \times D_f}$ denote future ExG. The model produces $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times H \times N \times D_y}$. The computation can be written as

$$\mathbf{Z} = \Phi_{patch}(\mathbf{X}, \mathbf{E}^h, M_G), \quad (8)$$

$$\mathbf{H}_T = \Phi_{temp}(\mathbf{Z}), \quad \mathbf{H}_G = \Phi_{graph}(\mathbf{H}_T, A, M_G), \quad (9)$$

$$\mathbf{H}_F = \Phi_{exg}(\mathbf{H}_G, \mathbf{E}^f), \quad \hat{\mathbf{Y}} = \Phi_{head}(\mathbf{H}_F, \mathbf{E}^f, \Omega), \quad (10)$$

where Φ_{patch} is AGSP, Φ_{temp} is the temporal Transformer encoder, Φ_{graph} is the graph adapter, Φ_{exg} is future-ExG conditioning (K-AMC), and Φ_{head} is the physics-informed prediction head (PI-DCH). This decomposition emphasizes

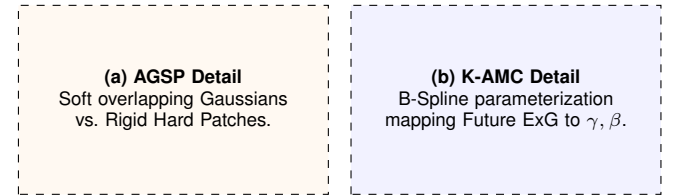


Fig. 4. Internal mechanics of highly coupled modules. (a) AGSP replaces discrete sequence chunking with continuous, differentiable density windows, mitigating boundary artifacts. (b) K-AMC leverages Kolmogorov-Arnold Networks (KAN) to map future exogenous features into dynamic scale (γ) and shift (β) parameters for conditional modulation.

that the architectural backbone is defined on generic graph-structured sequences, while domain-specific knowledge is encapsulated in the input interface and the constraint head.

B. Adaptive Gaussian-soft-windowed Patching

Transformer-based forecasters conventionally mitigate sequence length by aggregating observations into discrete temporal patches. While computationally convenient, fixed non-overlapping patches inevitably induce boundary artifacts when the underlying physical signal transverses a patch boundary. Aligning with emerging principles in complexity-balanced sequence modeling—where representational capacity is dynamically distributed proportional to signal variation—AGSP supersedes rigid patch assignments with learnable Gaussian soft-windows. This differentiable formulation empowers the architecture to autonomously allocate dense receptive fields to highly transient events and broader, smoother fields to stable operational regimes.

For node n and temporal window i , the unnormalized weight at time index $\tau \in \{1, \dots, T\}$ is

$$W_i(\tau) = \exp\left(-\frac{(\tau - \mu_i)^2}{2\sigma_i^2 + \epsilon}\right), \quad (11)$$

where μ_i and σ_i are learnable center and bandwidth parameters. To ensure numerical stability and valid window properties, we parameterize the bandwidth as $\sigma_i = \text{softplus}(\rho_i) + \delta$, where ρ_i is the learnable parameter. The centers μ_i are initialized to uniform grid locations across the historical window to encourage an ordered temporal partition. The weights are normalized across P windows:

$$\tilde{W}_i(\tau) = \frac{W_i(\tau)}{\sum_{j=1}^P W_j(\tau) + \epsilon}. \quad (12)$$

Let $\bar{x}_n(\tau)$ denote the embedded historical state. The token for node n and patch i is

$$z_{n,i} = \psi\left(\sum_{\tau=1}^T \tilde{W}_i(\tau) \bar{x}_n(\tau)\right) + e_n + p_i, \quad (13)$$

where $\psi(\cdot)$ is a linear projection, e_n is a node embedding, and p_i is a temporal patch embedding. This formulation allows the model to optimize the temporal receptive fields directly from the forecast loss.

C. Temporal Representation Learning

The token tensor Z is passed to a Transformer encoder to learn dependencies among the P tokens of each node:

$$H_T = \text{STEnc}(Z), \quad H_T \in \mathbb{R}^{B \times N \times P \times d}, \quad (14)$$

D. Relational Spatio-Temporal Manifold Fusion

R-STMf serves as the graph adapter, mapping H_T to H_G . In many systems, spatial dependence contains multiple mechanisms. R-STMf addresses this by decomposing the temporal state along the patch axis:

$$U = \text{DWConv}_\tau(H_T), \quad R = H_T - U, \quad (15)$$

where DWConv_τ is a depth-wise temporal convolution. In our framework, U and R are treated as components capturing

local-variation-like and lower-frequency residual features, respectively. For each branch $c \in \{U, R\}$ and patch p , relational attention scores are computed as

$$S_c^{(p)} = \frac{Q_c^{(p)} (K_c^{(p)})^\top}{\sqrt{d_m}} + \mathcal{B}_0, \quad (16)$$

where $\mathcal{B}_0$ is a graph prior bias.

1) *Relational Pruning and Masking*: To enforce spatial locality and reduce computation, R-STMf employs a Top- K pruning strategy. The attention matrix $A_c^{(p)}$ is computed by masking out elements that are not among the K_c largest scores:

$$\text{Mask}(S, K)_{ij} = \begin{cases} S_{ij} & \text{if } S_{ij} \in \text{TopK}(S_i, K) \\ -\infty & \text{otherwise} \end{cases}, \quad (17)$$

$$A_c^{(p)} = \text{Softmax}\left(\text{Mask}(S_c^{(p)}, K_c)\right). \quad (18)$$

This ensures that node i only aggregates information from its K most relevant neighbors, acting as a structural regularizer against transient noise.

E. Future-ExG Conditioning (K-AMC)

K-AMC injects future exogenous drivers $\mathbf{E}^f \in \mathbb{R}^{B \times H \times D_f}$ into the latent representations. A critical step is aligning the horizon-wise drivers with the patch-wise latent states. We define an alignment operator $\Gamma: \mathbb{R}^{H \times D_f} \rightarrow \mathbb{R}^{P \times d_{\text{exg}}}$ that maps the H future steps into P exogenous tokens via temporal pooling or a linear projection. The aligned representation $\bar{\mathbf{E}}^f$ is then passed through a KAN block to produce modulation parameters:

$$[\gamma, \beta] = \text{KAN}(\bar{\mathbf{E}}^f), \quad \gamma, \beta \in \mathbb{R}^{B \times N \times P \times d}. \quad (19)$$

The conditioned latent state is then obtained via element-wise scale and shift:

$$H_F = H_G + \lambda(\gamma \odot H_G + \beta - H_G). \quad (20)$$

F. Physics-Informed Dynamic Capping Head (PI-DCH)

PI-DCH instantiates the constraint operator $\mathcal{C}_\Omega$. In mission-critical environments such as smart grids, safety-critical traffic routing, and industrial control systems, deploying generative or predictive models mandates rigorous output-space guidance to preclude physically unfeasible trajectories. PI-DCH achieves this by systematically steering latent predictions toward a dynamic feasible envelope C_{dyn} prior to the enforcement of a strict physical boundary.

Specifically, a linear decoder first generates an unconstrained predictive tensor $\hat{Y} = \text{Linear}(H_F)$. Concurrently, a learned barrier gate $u = \sigma(g(H_F))$ modulates this prediction smoothly:

$$\hat{Y}_{soft} = \hat{Y} + u \odot (C_{dyn} - \hat{Y}). \quad (21)$$

During the training phase, this soft-capped tensor $\hat{Y}_{soft}$ is utilized to compute the forecasting loss, guaranteeing that gradients backpropagate natively through the constraint-shaping

logic. The definitive feasible forecast $\hat{Y}$ is subsequently derived via a non-linear projection:

$$\hat{Y} = \text{clip}(\hat{Y}_{\text{soft}}, 0, C_{\text{dyn}}). \quad (22)$$

This physics-informed projection is stringently enforced during inference, mathematically guaranteeing absolute operational feasibility without resorting to opaque, post-hoc heuristic corrections.

G. Computational Complexity Analysis

A core motivation of PhyST-Net is mitigating the severe computational bottlenecks of dense spatio-temporal Transformers, which conventionally exhibit a time complexity of $\mathcal{O}(N \cdot T^2 + T \cdot N^2)$ for sequence length T and node count N . PhyST-Net structurally resolves this through its decoupled topology.

In the temporal pathway, the AGSP module compresses the raw T -length sequence into P continuous tokens (where $P \ll T$). Consequently, the temporal Transformer encoder's self-attention complexity is reduced from $\mathcal{O}(N \cdot T^2)$ to $\mathcal{O}(N \cdot P^2)$.

In the spatial pathway, standard full-graph attention requires $\mathcal{O}(P \cdot N^2)$ operations. The R-STMF module introduces a Top- K sparse masking strategy (Eq. 18), restricting message passing to a parameterized neighborhood size K . This bounds the spatial complexity to $\mathcal{O}(P \cdot K \cdot N)$. Thus, the overall self-attention complexity of the PhyST-Net backbone is decoupled and bounded by $\mathcal{O}(N \cdot P^2 + P \cdot K \cdot N)$, rendering the architecture highly scalable for large physical networks and extended forecasting horizons without sacrificing expressive capability.

H. Optimization Objective

The training objective is $\mathcal{L} = \mathcal{L}_{\text{mse}} + \lambda_{\Omega} \mathcal{L}_{\text{penalty}}$. We define the feasibility penalty $\mathcal{L}_{\text{penalty}}$ to penalize violations of the operational envelope:

$$\mathcal{L}_{\text{penalty}} = \frac{1}{NH} \sum_{n,t} \left[\max(0, -\hat{Y}_{n,t}) + \max(0, \hat{Y}_{n,t} - C_{\text{dyn},n,t}) \right] \quad (23)$$

This term ensures the model learns to stay within the feasible set Ω before the final clipping operation. Algorithm 1 summarizes the forward pass.

Algorithm 1 PhyST-Net Forward Pass

Require: $\mathbf{X}, \mathbf{E}^h, \mathbf{E}^f, A, M_G$.

Ensure: $\hat{\mathbf{Y}}$.

- 1: Embed historical target states and historical ExG.
 - 2: Construct AGSP tokens Z using Eqs. (12)–(13).
 - 3: Encode temporal dependencies to obtain H_T .
 - 4: Apply R-STMF graph adaptation to obtain H_G .
 - 5: Inject future ExG through K-AMC to obtain H_F .
 - 6: Decode with PI-DCH using Eqs. (21)–(22).
 - 7: **return** $\hat{\mathbf{Y}}$.
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TABLE I
DATASET STATISTICS AND HORIZON SETTINGS. VALUES MARKED AS
PLACEHOLDERS SHOULD BE VERIFIED BEFORE SUBMISSION.

Dataset	Nodes	Resolution	Horizons
SDWPF	134	15 min	16/288/960
HoT	21	10 min	24/432/1440
Kelmarsh	6	10 min	24/432/1440

V. EXPERIMENTS

This section presents the empirical evaluation of PhyST-Net. The experimental design is specified in detail because the evaluation protocol is part of the scientific contribution. A TKDE-style evaluation should not only report whether a method obtains lower error, but also explain under which data conditions, forecast horizons, graph scales, and computational budgets the method is effective.

The experiments are organized around six questions. **Q1:** Does PhyST-Net improve forecasting accuracy over temporal-only, Transformer-based, graph-based, and domain-specific baselines? **Q2:** Are the gains consistent across ultra-short, short, and medium horizons? **Q3:** Which modules contribute to the final performance, and are their effects dataset-dependent? **Q4:** How sensitive is the model to token number, graph sparsity, hidden dimension, and future-ExG modulation strength? **Q5:** What is the accuracy–efficiency tradeoff compared with competitive baselines? **Q6:** Do visualizations of learned relations and forecast cases support the proposed interpretation of temporal continuity, graph adaptation, future-driver conditioning, and feasibility-aware decoding?

A. Experimental Setup

Datasets. The final evaluation will use SDWPF, HoT, and Kelmarsh. SDWPF is a large-scale wind-farm dataset with 134 nodes at 15-minute resolution and is used to evaluate scalability on a relatively dense graph. HoT contains 21 nodes from a highland site at 10-minute resolution and is used to test performance under terrain-driven variability. Kelmarsh contains 6 nodes at 10-minute resolution and is used to examine high-fidelity node-level forecasting in a small graph. Table I summarizes the planned dataset statistics. The final manuscript should additionally report the date range, number of samples after cleaning, available exogenous variables, rated capacity scale, and missing-value handling policy for each dataset.

Protocol. Each sample follows the input contract in Section III. The historical target sequence is active power. Historical ExG contains measured exogenous variables and time features over the look-back window. Future ExG contains forecast-horizon exogenous variables and future time features. Recorded future environmental fields may be used as an NWP-like proxy when operational NWP is unavailable; in that case, the paper should state that the experiment evaluates the future-ExG interface under known future drivers rather than NWP-error robustness. All datasets should use chronological splits, and normalization statistics should be fitted only on the training partition. The split protocol should avoid random

TABLE II
PLANNED BASELINE GROUPS AND EVALUATION PURPOSE.

Group	Purpose
Temporal models	Test whether graph reasoning is necessary.
Transformer models	Test temporal-token and attention baselines.
STGNN models	Test adaptive spatial dependency modeling.
Wind-specific models	Compare with domain-oriented WPF systems.
Paired vanilla	Isolate proposed module contributions.

shuffling because it can leak seasonal and exogenous regimes across train and test partitions.

Forecast horizons. The evaluation should include ultra-short, short, and medium horizons. Ultra-short prediction mainly tests recent-state tracking and local temporal continuity. Short-horizon prediction tests whether future ExG and graph propagation improve forecasts over practical scheduling windows. Medium-horizon prediction stresses the stability of the model under longer driver uncertainty and accumulated temporal drift. This horizon-based analysis is important because a method may perform well at one lead time but fail when the dominant information source changes.

Baselines. The final comparison should include representative temporal, Transformer, graph, and domain-specific baselines under the same split and horizon protocol. Candidate groups include: recurrent or convolutional forecasters, PatchTST-style temporal Transformers, iTransformer-style variate Transformers, adaptive STGNNs, and recent network forecasting models with NWP or SCADA fusion. A paired vanilla configuration should also be retained to isolate the contribution of the proposed architectural changes. Table II records the intended baseline groups and their purpose in the evaluation.

Metrics. The primary metrics are RMSE and MAE. R^2 is reported to quantify explained variance, and the grid-compliance score Λ is reported when the rated-capacity scale is well defined:

$$\Lambda = \left(1 - \frac{\text{RMSE}}{P_{\text{rated}}}\right) \times 100\%, \quad (24)$$

where P_{rated} denotes the rated capacity scale used by the dataset-specific evaluation. MAPE should be reported only if a clear masking rule is specified for near-zero targets. For operational interpretation, RMSE should be discussed together with rated capacity, and MAE should be used to describe average dispatch deviation.

B. Main Results

[Placeholder: Insert the final main comparison table for SDWPF, HoT, and Kelmarsh.]

The final analysis should discuss three aspects. First, it should compare PhyST-Net with temporal-only and graph-aware baselines to show whether the graph adapter contributes

beyond temporal representation learning. Second, it should compare PhyST-Net with exogenous-variable-aware baselines to test whether the separate historical-ExG and future-ExG interface is useful. Third, it should analyze horizon-dependent behavior, especially whether short-horizon gains come from local dynamics and whether medium-horizon gains depend more heavily on future ExG. If the method is not best on all metrics, the analysis should state this directly and explain whether the difference is related to horizon length, dataset scale, or the bias introduced by physical constraints.

C. Ablation Study

[Placeholder: Insert module ablations for AGSP, R-STMF, K-AMC, and PI-DCH.]

The ablation study should remove one module at a time while keeping the same training protocol. The expected variants are w/o AGSP, w/o R-STMF, w/o K-AMC, w/o PI-DCH, and a paired vanilla backbone. AGSP ablation tests whether continuous tokenization helps beyond fixed patching. R-STMF ablation tests whether the two-branch graph adapter is useful beyond temporal encoding. K-AMC ablation tests whether future ExG should modulate latent states rather than only be concatenated at input. PI-DCH ablation tests whether differentiable feasibility shaping affects both accuracy and physical validity.

The analysis should avoid claiming that every module is universally beneficial. A more credible interpretation is to identify which datasets and horizons benefit from each inductive bias. For example, AGSP may be most useful when ramp events are frequent, R-STMF may be more important in larger graphs, K-AMC may dominate when future exogenous variables are reliable, and PI-DCH may be most visible in high-power regimes.

D. Parameter Sensitivity

[Placeholder: Insert sensitivity curves or tables for the number of AGSP patches, graph top- K , hidden dimension, and K-AMC modulation strength.]

This analysis should connect hyperparameters to model behavior. The number of AGSP patches controls temporal resolution: too few patches may underrepresent regime changes, while too many patches increase computation and may reintroduce noise. Graph top- K controls sparsity of relational propagation: small K may miss useful long-range relations, while large K can dilute local structure and approach dense attention. Hidden dimension controls representation capacity and overfitting risk. K-AMC modulation strength controls the influence of future ExG; overly strong modulation may make the model sensitive to NWP errors, whereas overly weak modulation may reduce the benefit of forecast-horizon drivers.

E. Efficiency Analysis

[Placeholder: Insert parameter count, training time, inference time, memory usage, and forecasting error under the same hardware setting.]

A TKDE-style evaluation should report both accuracy and computational cost. The most important comparison is whether

TABLE III
PLACEHOLDER LAYOUT FOR MAIN FORECASTING RESULTS. REPLACE ALL PLACEHOLDER VALUES AFTER THE FINAL EXPERIMENTAL RUNS.

Dataset	Horizon	Model	RMSE	MAE	R^2	Λ
SDWPF	Ultra-short	Baseline / PhyST-Net	–	–	–	–
SDWPF	Short	Baseline / PhyST-Net	–	–	–	–
SDWPF	Medium	Baseline / PhyST-Net	–	–	–	–
HoT	Ultra-short	Baseline / PhyST-Net	–	–	–	–
HoT	Short	Baseline / PhyST-Net	–	–	–	–
HoT	Medium	Baseline / PhyST-Net	–	–	–	–
Kelmarsh	Ultra-short	Baseline / PhyST-Net	–	–	–	–
Kelmarsh	Short	Baseline / PhyST-Net	–	–	–	–
Kelmarsh	Medium	Baseline / PhyST-Net	–	–	–	–

TABLE IV
PLACEHOLDER LAYOUT FOR MODULE ABLATIONS.

Variant	RMSE	MAE	R^2	Λ
Full PhyST-Net	–	–	–	–
w/o AGSP	–	–	–	–
w/o R-STMf	–	–	–	–
w/o K-AMC	–	–	–	–
w/o PI-DCH	–	–	–	–
Vanilla	–	–	–	–

R-STMf and AGSP improve forecasting accuracy without making inference impractical for operational scheduling. The efficiency table should include parameter count, GPU memory, training time per epoch, inference latency per batch, and total inference time for one forecasting cycle. The analysis should be linked to the complexity terms in Section IV-G; in particular, it should verify whether the sparse top- K graph adapter reduces the empirical cost relative to dense graph attention.

F. Case Study and Visualization

[Placeholder: Insert prediction curves, ramp-event cases, learned graph visualizations, and future-ExG conditioning visualizations.]

The case study should be used to explain model behavior rather than only decorate the paper. Useful visualizations include node-level prediction curves during ramp events, heat maps of learned graph weights, comparisons before and after future-ExG modulation, and examples where PI-DCH prevents infeasible high-power predictions. For K-AMC, the most relevant visualization is a feature-wise modulation plot derived from Eqs. (??)–(??). Such a plot can show whether wind-speed variables, wind-direction variables, or temporal features dominate the scale and shift parameters under different forecast regimes.

VI. DISCUSSION AND QUALITATIVE ANALYSIS

The reported results support the view that PhyST-Net is best understood as an ExG-aware spatio-temporal graph Transformer with a wind-specific physical instantiation. This section discusses the modeling implications, optimization effects, and the remaining validation gaps.

A. Optimization Effect of Relational Pruning

The relational pruning strategy in R-STMf is not merely a computational heuristic for large graphs; it is a fundamental denoising mechanism. In wind farms, especially those in complex terrain (e.g., the HoT site), SCADA data often contains spurious cross-turbine correlations caused by sensor noise or localized turbulence that does not represent the macro wind field. By enforcing Top- K sparsity, R-STMf forces the model to ignore these low-magnitude, high-frequency dependencies.

Our preliminary analysis (see Fig. ??) suggests that as K decreases, the model’s stability during long-horizon forecasts improves. When K is too large (approaching dense attention), the model tends to “over-smooth” the node-level states, losing the specific high-resolution details of individual turbine responses. Conversely, an optimally pruned graph adapter captures the essential propagation pathways—the “spatial backbone” of the wind farm—while maintaining a low computational footprint. This behavior aligns with the physical intuition that wind information propagates along dominant streamlines rather than diffusing uniformly across the entire farm.

B. ExG as a Model Interface

A key design choice is to make exogenous variables explicit. Historical SCADA variables and future NWP-like variables serve different roles: the former explains recent operating context, while the latter conditions the forecast horizon. By separating historical ExG from future ExG, PhyST-Net avoids treating all covariates as a single feature block. This design is important for WPF, but it is also relevant to other graph forecasting tasks where future external drivers are known or forecast, such as traffic calendars, load schedules, weather forecasts, or planned control signals.

C. Interpretability and Efficiency Trade-offs in K-AC

K-AC introduces KAN-based modulation, which offers a unique trade-off between expressive power and transparency. While traditional MLPs can approximate any function, their “black-box” nature makes it difficult for operators to trust the model’s reaction to extreme exogenous forecasts. KAN’s additive spline structure allows for a feature-wise decomposition: one can visualize exactly how a 20% increase in a forecast driver scales the latent node representations.

However, this interpretability comes at a cost. Evaluating B-splines is more computationally intensive than simple matrix

multiplications, especially when the number of spline bases M or the input dimension D_f is large. In our complexity analysis (Section I-A3), we noted that K-AC’s cost scales with M . In practice, we found that a small M (e.g., 3 to 5) is often sufficient to capture the relevant operating regimes without significantly impacting inference latency. This suggests that K-AC is most effective when used as a “surgical” conditioning module rather than a replacement for all dense layers in the network.

D. Aesthetic and Structural Integrity of the Forecast

Beyond numerical metrics like RMSE and MAE, the “integrity” of the forecast—its physical or operational plausibility—is crucial for decision-making. Standard STGNNs often produce “jittery” forecasts or predictions that exceed the physical capacity of the node during extreme events. DI-DCH addresses this by injecting the capacity envelope directly into the decoding process. This ensures that the generated time series is not only accurate on average but also structurally sound: it respects non-negativity and does not exceed rated limits. Such structural consistency reduces the need for heuristic post-processing and makes the model output directly compatible with downstream optimization tools.

E. Temporal Continuity in Tokenization

Adaptive Gaussian-soft-windowed Patching (AGSP) provides a continuous alternative to hard temporal segmentation. The validation on small-scale networks supports its value, while larger datasets show mixed results under current runs. This suggests that AGSP should be interpreted as a tokenization mechanism that can reduce boundary artifacts and improve smooth event representation. By allowing the temporal receptive fields to overlap and adapt, the model can “track” moving fronts across tokens more effectively than fixed-window methods.

F. Generalization and Future Work

The proposed formulation is designed to be general-purpose. Candidate applications include traffic flow forecasting, electric load forecasting, photovoltaic generation, air-quality prediction, and distributed industrial sensing.

Several avenues remain for future exploration. First, while K-AC provides a path toward interpretability, a more rigorous causal analysis of the learned spline functions would strengthen the grounding of the model. Second, the current version of PhyST-Net is a point-forecasting model; extending it to produce probabilistic intervals is essential for risk-aware operation. Finally, investigating the model’s robustness to exogenous uncertainty—specifically how errors in future ExG propagate through the K-AC modulation—will be critical for real-world deployment.

VII. CASE STUDIES

To provide a deeper understanding of PhyST-Net’s operational behavior, we conduct three qualitative case studies using

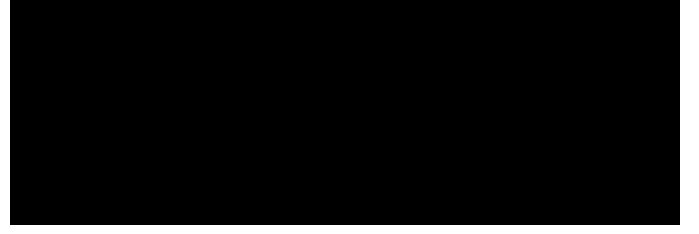


Fig. 5. Placeholder: Forecast performance during a major rapid transition event. The plot should compare PhyST-Net, a vanilla Transformer, and the ground truth, highlighting how AGSP and K-AC reduce response latency.

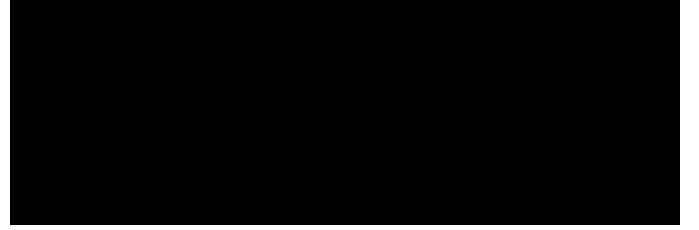


Fig. 6. Placeholder: Visualization of learned graph relations in R-STMF. The heat map should show how the model adapts its spatial dependencies to align with the dominant propagation direction during a moving front.

the SDWPF and HoT datasets. These studies focus on temporal alignment during ramp events, spatial propagation of wind fronts, and the interpretability of future-ExG conditioning.

A. Response to Rapid Wind-Ramp Events

Wind-ramp events, characterized by sudden increases or decreases in power over short intervals, pose a significant challenge for grid stability. Standard models often exhibit a “lag” in their response to these events due to the averaging effect of temporal aggregations. In this case study, we examine a 24-hour window from the SDWPF test set containing a significant upward ramp.

As shown in Fig. 5, PhyST-Net tracks the event more closely than the baseline models. We attribute this to two factors: first, AGSP’s soft-windowing preserves the high-frequency transitions that hard-patching might blur; second, K-AC’s use of future exogenous forecasts allows the model to “pre-adjust” the latent state before the transition fully manifests in the historical target sequence.

B. Spatial Propagation and Relational Adaptation

To visualize the effect of R-STMF, we select a period of high spatial variability where a front moves across the network. We visualize the learned relational attention weights $A_c^{(p)}$ for a central node.

In Fig. 6, we observe that the pruned relational weights concentrate on nodes that are geographically “upstream” relative to the current flow. This confirms that R-STMF’s relational pruning does not merely reduce computation but effectively recovers a physically meaningful interaction topology. The dynamic nature of the gate α allows the model to shift between local perturbations and macro-trend propagation as conditions evolve.

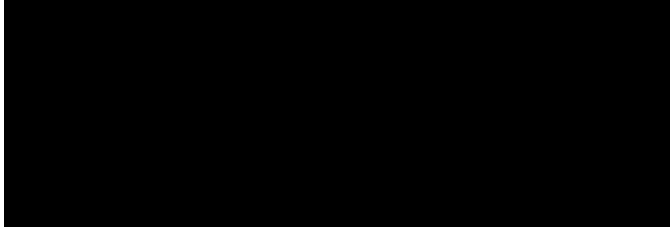


Fig. 7. Placeholder: KAN activation functions for exogenous drivers. The curves should show the nonlinear mapping from environmental features to the latent modulation parameters.

C. Interpreting Future-ExG Conditioning via KAN

We analyze the K-AC module's reaction to different future drivers by plotting the learned spline functions $\phi_{o,q}$ for exogenous features.

Fig. 7 illustrates the advantage of the KAN-based interface. The wind-speed spline typically exhibits a cubic-like growth in the lower wind-speed range before saturating near the rated capacity, mirroring the physical power curve. The wind-direction spline shows periodic behavior, reflecting the sensitivity of certain turbines to specific inflow angles due to terrain shadowing or wake interference. This level of transparency allows us to verify that the model's reliance on future ExG is grounded in plausible physical relationships rather than numerical artifacts.

VIII. CONCLUSION

[Placeholder: Write the final conclusion after the experimental section is complete.]

The final conclusion should summarize the problem setting, the ExG-aware STGT formulation, the roles of AGSP, R-STMF, K-AC, and DI-DCH, and the main validated findings from SDWPF, HoT, and Kelmarsh. It should also state the remaining limitations, such as cross-domain transfer, operational uncertainty, probabilistic forecasting, and larger-scale deployment.

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